REPORT LAB 5:

MEASURING THE VISCOSITY OF A LIQUID USING STOKES'S METHOD

Class: ..CC10.. / Group: ..01..Team:…….	Lecturer's comment
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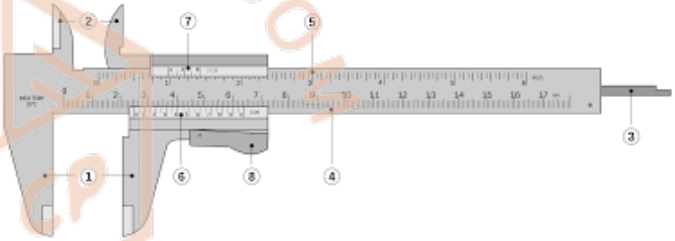
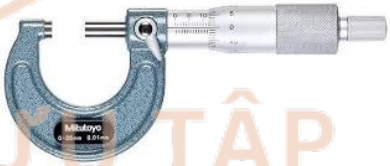
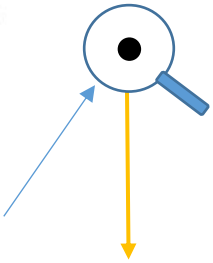
I. Aims/Purposes

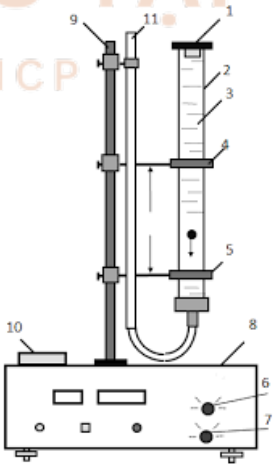
- To study the internal friction phenomena liquid.
- To determine the coefficient of viscosity of liquid by stoke's method.

II. Apparatus, Methods, and Procedure









- Apparatus: **MN-971A** include:

- Glass tubes 95cm high, carved of 2mm / line
- Lubricant with viscosity coefficient
- Steel balls
- Hopper oriented the dropping steel balls
- Small Magnets to get the steel balls out of liquids
- Device to measure the fall time of steel balls
- Two sensors
- Vernier caliper 0-25mm, accuracy 0.02 mm.
- Chronometer, accuracy 0.001s.
- Micrometer, accuracy 0.01 mm.
- Thermometer, accuracy 1°.
- Technical balance, accuracy 0.02 g.

- Methods:

- When a metallic ball is allowed to fall through a liquid, where its weight is balanced by the viscous(drag). Stokes's tore due to the viscosity of the fluid and botany tore acting on it.

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- Procedure:

1. Measure **the diameter d** of steel balls by Micrometer.
2. Measure **the time interval t** of steel balls falling in liquids.

2.1	Installation and adjustment of balance
2.2	Adjusting the sensitivity of the sensor 4 and 5 of the time measurement
2.3	Measurement falling time of steel balls

3. Read and record the following data in the table.

III. Equations

Calculate the absolute error of d, m, t, L:

$$\Delta d = \Delta d_{sys} + \overline{\Delta d}$$
$$\Delta m = \Delta m_{sys} + \overline{\Delta m}$$
$$\Delta t = \Delta t_{sys} + \overline{\Delta t}$$
$$\Delta L = \Delta L_{sys} + \overline{\Delta L}$$

Calculate the density of the steel ball:

$$\bar{\rho} = \frac{\bar{m}}{\frac{1}{6}\pi \bar{d}^3}$$

$$\frac{\Delta\rho}{\bar{\rho}} = \frac{\Delta\pi}{\pi} + \frac{\Delta m}{\bar{m}} + 3\frac{\Delta d}{\bar{d}}$$

Calculate the viscosity of the liquid:

$$\eta = \frac{1}{18}\frac{(\rho - \sigma)d^2 g t}{L\left(1 + 2.4\frac{d}{D}\right)}$$

WHERE:

η - Coefficient of viscosity of liquid *(kg m^{-1} s^{-1})*

$r \rightarrow$ radius of spherical ball *(m)*

$p \rightarrow$ density of the steel ball (*kg m^{-3}*)

$\sigma \rightarrow$ density of the liquid (*kg m^{-3}*)

$g \rightarrow$ acceleration due to gravity *(9.8 m s^{-2})*

$D \rightarrow$ diameter of the glass cylinder *(mm)*

$d \rightarrow$ diameter of steel ball *(mm)*

$t \rightarrow$ time *(s)*

$$\frac{\Delta\eta}{\bar{\eta}} = \frac{\Delta\rho + \Delta\sigma}{\bar{\rho} - \bar{\sigma}} + \frac{\Delta g}{g} + \frac{\Delta t}{\bar{t}} + \frac{\Delta L}{\bar{L}} + \frac{1}{\bar{D} + 2.4\bar{d}}\left[\left(2\bar{D} + 2.4\bar{d}\right)\frac{\Delta d}{\bar{d}} + 2.4\bar{d}\frac{\Delta D}{\bar{D}}\right]$$

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Result:

$$\rho = \bar{\rho} \pm \Delta\rho \left(\frac{kg}{m^3}\right)$$

$$\eta = \bar{\eta} \pm (s_t \Delta\eta) \left(\frac{kg}{ms}\right)$$

IV. Experimental data

Table 1

- Accuracy of micrometer: 0.01 (*mm*) **- Accuracy of chronometer:** 0.001 (*sec*) **- Diameter of cylinder:** D = 34.82 ± 0.02 (*mm*) **- Room temperature:** T°C= 30 ± 1 (°C) **- Gravitational acceleration:** g = 9.810 ± 0.005 (m/s^2) **- Constant Pi:** π = 3.140 ; $\Delta\pi$ = 0.005	**- Density of steel ball:** ρ = 7894.8014 ± 236.0546 (*kg/m*^{3}) **- Mass of steel ball:** m = 1.04 ± 0.02 (*g*) **- Density of lubricant oil depended on the room temperature:** $\sigma = 895 \pm 89$ (*kg/m*^{3}) **- Distance between two sensors:** L = 300 ± 1 (*mm*)

	d (*mm*)	*Δd* (*mm*)	*t* (*sec*)	*Δt* (*sec*)
1	6.31	0.004	0.978	0.0102
2	6.32	0.006	0.970	0.0022
3	6.31	0.004	0.955	0.0128
4	6.33	0.016	0.957	0.0108
5	6.30	0.014	0.979	0.0112
Avg.	6.314	0.009	0.9678	0.0094

V. Calculations

The absolute error of d, m, t, L:

$$\Delta d = \Delta d_{sys} + \overline{\Delta d} = 0.01 + 0.009 = 0.019\ (10^{-3}m)$$

$$\Delta m = \Delta m_{sys} + \overline{\Delta m} = 0.02 + 0 = 0.02\ (10^{-3}kg)$$

$$\Delta t = \Delta t_{sys} + \overline{\Delta t} = 0.001 + 0.0094 = 0.0104\ (sec)$$

$$\Delta L = \Delta L_{sys} + \overline{\Delta L} = 1 + 0 = 1\ (10^{-3}m)$$

The density of the steel ball:

$$\bar{\rho} = \frac{\bar{m}}{\frac{1}{6}\pi\bar{d}^3} = \frac{1.04\text{x}10^{-3}}{\frac{1}{6}\text{ x } 3.140 \text{ x } (6.314\text{x}10^{-3})^3} \approx 7894.8014 \left(\frac{kg}{m^3}\right)$$

$$\frac{\Delta\rho}{\bar{\rho}} = \frac{\Delta\pi}{\pi} + \frac{\Delta m}{\bar{m}} + 3\frac{\Delta d}{\bar{d}} = \frac{0.005}{3.140} + \frac{0.02}{1.04} + \frac{3 \text{ x } 0.019\text{x}10^{-3}}{6.314\text{x}10^{-3}} \approx 0.0299$$

$$=> \ \Delta\rho \approx 236.0546 \left(\frac{kg}{m^3}\right)$$

The viscosity of the liquid:

$$\bar{\eta} = \frac{1}{18}\frac{(\bar{\rho} - \bar{\sigma})\bar{d}^2\bar{g}\bar{t}}{\bar{L}\left(1 + 2.4\frac{\bar{d}}{\bar{D}}\right)} = \frac{(7894.8014 - 895)\text{ x }(6.314\text{x}10^{-3})^2\text{ x }9.810\text{ x }0.9678}{18\text{ x }300\text{ x }\left(1 + 2.4\text{ x }\frac{6.314\text{x}10^{-3}}{34.82\text{x}10^{-3}}\right)} \approx 0.3419\left(\frac{kg}{ms}\right)$$

$$\frac{\Delta\eta}{\bar{\eta}} = \frac{\Delta\rho + \Delta\sigma}{\bar{\rho} - \bar{\sigma}} + \frac{\Delta g}{g} + \frac{\Delta t}{\bar{t}} + \frac{\Delta L}{\bar{L}} + \frac{1}{\bar{D} + 2.4\bar{d}}\left[\left(2\bar{D} + 2.4\bar{d}\right)\frac{\Delta d}{\bar{d}} + 2.4\bar{d}\frac{\Delta D}{\bar{D}}\right]$$

$$= \frac{236.0546 + 89}{7894.8014 - 895} + \frac{0.005}{9.810} + \frac{0.0104}{0.9678} + \frac{0.001}{0.300}$$

$$+ \frac{1}{34.82\text{x}10^{-3} + 2.4\text{ x }6.314\text{x}10^{-3}}\left[(2\text{ x }34.82\text{x}10^{-3}\right.$$

$$\left. + 2.4\text{ x }6.314\text{x}10^{-3})\frac{0.019\text{x}10^{-3}}{6.314\text{x}10^{-3}} + 2.4\text{ x }6.314\text{x}10^{-3}\text{ x }\frac{0.02\text{x}10^{-3}}{34.82\text{x}10^{-3}}\right]$$

$$\approx 0.0663\left(\frac{kg}{ms}\right)$$

$$=> \ \Delta\eta \approx 0.0227\left(\frac{kg}{ms}\right)$$

Conclusion:

$$\eta = (\bar{\eta} \pm \Delta\eta)\left(\frac{kg}{ms}\right) = (0.3419 \pm 0.0227)\left(\frac{kg}{ms}\right)$$

Extra work: Using t-Student distribution for calculating the uncertainty

$$\Delta\eta = \pm\sqrt{\frac{\sum_{i=1}^{n}\Delta\eta_i^2}{n(n-1)}}, \qquad n = 10$$

The correction with Student coefficient, s_t for $\alpha = 0.95$ and $n = 10$ has to be done and the final results should be written as $\eta = \bar{\eta} \pm (s_t\Delta\eta)\left(\frac{kg}{ms}\right)$

VI. Question

1. Explain the emergence of internal friction, stating the basic theory and written expression of this force. The unit of fluid viscosity coefficient.

- When a fluid actually flows between layers with different velocities, frictional forces will arise, there are forces between the layers of moving fluid and these are called internal frictional forces. The properties of fluids that are related to the presence of internal friction are called viscosities.

- Thus, the lower layer exerts a frictional force on the layer above it, which slows down the movement of the upper layer, and conversely, the upper layer exerts an accelerating force on the layer below it.The internal friction force will decrease as the temperature increases.

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Where:

- F_{MSN}: internal friction(N)
- S: contact area(m^2)
- η: ccoefficient of viscosity of liquid*(kg/m.s)*
- △v: velocity difference of the surfaces
- △h: distance between surfaces

$$F_{MSN} = S\eta \frac{\Delta v}{\Delta h}$$

2. Presentation Stokes method of determination of fluid viscosity. Explain the causes and characteristic of force against the motion of the steel balls fall in the liquid.

- The viscosity coefficient can determine by the Stokes method through the device MN-971A.
- The viscosity of a liquid is determined by the Stokes method through the following formulas.
  - When the steel balls with mass m drop through the hopper than falls in the liquid, the marble are affected by three forces:
  - ✓ Gravity P is directed vertically from the top and has value equals:

$$P = mg = \frac{4}{3}\pi.r^3\rho_1.g$$

With r is the radius and ρ 1 bulk density of steel balls, g is the gravitational acceleration.

- ✓ Acsimet force F_A is vertical direction from the bottom up and has value by the weight of the liquid being occupied by the steel balls:

$$F_A = \frac{4}{3}\pi r^3.\rho.g$$

With ρ is bulk density of liquid

- ✓ The internal friction Force F_C is vertically from the bottom up and have value by:

$$F_C = 6\pi.\eta.r.v$$

With v is velocity and η is viscosity coefficient of liquid

- Under the effect of the forces mentioned above, marble will move with acceleration $a = \frac{dv}{dt}$ following the second Newton laws:

$$m\frac{\overrightarrow{dv}}{dt} = \vec{P} + \overrightarrow{F_A} + \overrightarrow{F_C} \quad (1)$$

- The acceleration a make the velocity v of steel balls gradually increase, on the other hand when v increases, internal friction increases. When V reaches the value vo, the Acsimet force and internal friction will completely eliminate gravity P, steel balls will reach uniform speed motion.

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- Give equation (1) is 0 and follow the direction of steel balls motion, we have:

$$\frac{4}{3}\pi.r^3\rho_1.g - \frac{4}{3}\pi r^3.\rho.g - 6\pi.\eta.r.v_0 = 0$$

$$\eta = \frac{2}{9}\frac{(\rho_1 - \rho_o).r^2.g}{v_o} \qquad (2)$$

*v_0 can be determined by measuring the number of time interval movement of steel balls that are falling between two straight baseline 4 and 5 separated by a distance L.

$$v_0 = \frac{L}{\tau}$$

- Substitute v_0 in (2) with d is the diameter of the steel balls, we find:

$$\eta = \frac{1}{18}\frac{(\rho_1 - \rho).d^2.g.\tau}{L} \qquad (3)$$

- In fact, liquid is not infinitely wide, it is contained in a cylinder with a finite diameter d. In this case, the viscosity coefficient of the fluid is calculated using the formula:

$$\eta = \frac{1}{18}\frac{(\rho_1 - \rho).d^2.g.\tau}{L.(1 + 2,4\frac{d}{D})} \qquad (4)$$

* If you know ρ_1, ρ, g, L, D the viscosity coefficient of the liquid can determine in a simple way, by measuring the diameter d of steel balls and period τ were between two selected baseline.

3. How the speed of the steel balls falls in liquid change? Why the fall time measurements of the steel balls are made at the end of the glass tube?

- Initially, when you drop it in, you will see a speed change. If dropped gently from the open surface, the ball will accelerate. But gradually increasing to a certain threshold does not increase anymore, it falls evenly! And if dropped a distance above the open surface, after free fall in the air, the marble plunges into liquid with some speed. But it eventually settles down to a certain drop rate.

- The initial velocity is not stable, so the measurement of the falling time of the ball is made at the end of the glass tube.

4. Under what conditions, we can calculate the viscosity coefficient of fluid by the formula (3) or formula (4)?

- v_0 can be determined by measuring the number of time interval movement of steel balls that are falling between two straight baseline 4 and 5 separated by a distance L.

$$\eta = \frac{1}{18}\frac{(\rho_1 - \rho).d^2.g.\tau}{L}$$

- In the case of quality in a pipe with an inner diameter of d, then:

$$\eta = \frac{1}{18}\frac{(\rho_1 - \rho).d^2.g.\tau}{L.(1+2,4\frac{d}{D})}$$

5. Apply formulas following, demonstrated relative error of viscosity coefficient.

$$\frac{\Delta\eta}{\eta} = \frac{\Delta\rho_1 + \Delta\rho}{\rho_1 - \rho} + \frac{\Delta g}{g} + \frac{\Delta\tau}{\tau} + \frac{\Delta L}{L} + \frac{1}{D+2,4d}\left[(2D+2,4d)\frac{\Delta d}{d} + 2,4d\frac{\Delta D}{D}\right]$$

- Set up the formula for calculating the viscosity error

$$\eta = \frac{1}{18}\frac{(\rho_1 - \rho).d^2.g.\tau}{L.(1+2,4\frac{d}{D})}$$

- Step 1: Logarithm of nepal on both sides

$$ln\eta = ln\left[\frac{(\rho_1 - \rho)d^2 g\tau}{18L\left(1 + 2.4\frac{d}{D}\right)}\right] = ln(\rho_1 - \rho) + 2lnd + lng + ln\tau - ln18 - lnL - ln\left(1 + 2.4\frac{d}{D}\right)$$

- Step 2: Completely Differentiate Both Sides

$$d(ln\eta) = d\left[ln(\rho_1 - \rho) + 2lnd + lng + ln\tau - ln18 - lnL - ln\left(1 + 2.4\frac{d}{D}\right)\right]$$

- Step 3: Shortened Transform

$$\frac{d\eta}{\eta} = \frac{d(\rho_1 - \rho)}{(\rho_1 - \rho)} + 2\frac{dd}{d} + \frac{dg}{g} + \frac{d\tau}{\tau} - \frac{dL}{L} - \frac{d\left(1 + 2,4\frac{d}{D}\right)}{\left(1 + 2,4\frac{d}{D}\right)}$$

$$\frac{d\eta}{\eta} = \frac{d\rho_1 - d\rho}{(\rho_1 - \rho)} + 2\frac{dd}{d} + \frac{dg}{g} + \frac{d\tau}{\tau} - \frac{dL}{L} - \frac{2,4\frac{dd}{D} - 2,4\frac{d.dD}{D^2}}{\left(1 + 2,4\frac{d}{D}\right)}$$

$$\frac{d\eta}{\eta} = \frac{d\rho_1 - d\rho}{(\rho_1 - \rho)} + 2\frac{\mathbf{dd}}{\mathbf{d}} + \frac{dg}{g} + \frac{d\tau}{\tau} - \frac{dL}{L} - \frac{\mathbf{2,4dd}}{\mathbf{(D + 2.4d)}} + 2,4\frac{d.dD}{D(D + 2.4d)}$$

$$\frac{d\eta}{\eta} = \frac{d\rho_1 - d\rho}{(\rho_1 - \rho)} + \frac{dg}{g} + \frac{d\tau}{\tau} - \frac{dL}{L} + \frac{\mathbf{(2D + 2,4d)dd}}{\mathbf{d.(D + 2.4d)}} + 2,4\frac{d.dD}{D(D + 2.4d)}$$

$$\frac{d\eta}{\eta} = \frac{d\rho_1 - d\rho}{(\rho_1 - \rho)} + \frac{dg}{g} + \frac{d\tau}{\tau} - \frac{dL}{L} + \frac{1}{(D + 2.4d)}\left[\frac{(2D + 2,4d)dd}{d} + 2,4\frac{d.dD}{D}\right]$$

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